

3) Qualitative Data Collection Techniques

Training Course in Sexual and
Reproductive Health Research

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Objective of Presentation

By the end of this presentation you should be able to:

- Describe different types of data collection techniques
- Demonstrate dimensions , type of observations and how to prepare and conduct observation
- Understand the practical communication skills for interviews to ask good questions , probe and follow up questions
- Able to prepare for interview
- Understand the characteristics and uses of focus group discussions
- Conduct focus group discussions

Overview of Data Collection Techniques



- Data collection techniques allow us to systematically collect information about our objects of study (people, objects, phenomena) and about the settings in which they occur.
- In the collection of data we have to be systematic. If data are collected haphazardly, it will be difficult to answer our research questions in a conclusive way.

Data Collection Techniques

The logo of the Punjab Veterinary and Animal Sciences University (PUACP) is centered in the background. It features a shield with a yellow border, containing a blue and white emblem of an open book and a banner with Urdu text. The acronym 'PUACP' is written in blue at the bottom of the shield.

1. Document Review
2. Observation
3. Interview (face-to-face)
4. Focus Group Discussion

Document Review



1. Document Review

- A qualitative research project may require review of documents such as:
 - Course syllabi
 - Faculty journals
 - Meeting minutes
 - Strategic plans
 - Newspapers

1. Document Review

- Depending on the research question, the researcher might utilize:
 - Rating scale
 - Checklist
 - Content analysis
 - Matrix analysis



Observation

2. Observation

- OBSERVATION is a technique that involves systematically selecting, watching and recording behaviour and characteristics of living beings, objects or phenomena.
- Without training, our observations will heavily reflect our personal choices of what to focus on and what to remember.
- You need to heighten your sensitivity to details that you would normally ignore and at the same time to be able to focus on phenomena of true interest to your study.

Type of Observation

Observation of human behaviour

- **Participant observation:** The observer takes part in the situation he or she observes
 - Example: a doctor hospitalised with a broken hip, who now observes hospital procedures ‘from within’
- **Non-participant observation:** The observer watches the situation, openly or concealed, but does not participate

Type of Observation

- **Open**
 - (e.g., 'shadowing' a health worker with his/her permission during routine activities)
- **Concealed**
 - (e.g., 'mystery clients' trying to obtain antibiotics without medical prescription)
- **Observations of objects**
 - For example, the presence or absence of a latrine and its state of cleanliness

Type of Observation

1. **General observation** may be used as the starting point in to be familiar with the setting and the new context.
2. **More focus observation** may be used to evaluate whether people really do what they say they do.
3. **Access the unspoken knowledge of subject**, that is, the subconscious knowledge that they would not be able to verbalise in an interview setting.
4. **Compare a phenomena** and its specific components in greater detail.

Dimensions of Observation

1. **Space** (physical places)
2. **Actors** (people involved)
3. **Activities** (the set of related acts people do)
4. **Object** (the physical things that are present)
5. **Time** (the sequencing that takes place over time)
6. **Goal** (the things people are trying to accomplish)
7. **Feeling** (the emotions felt and expressed)

(Spradlet 1979)

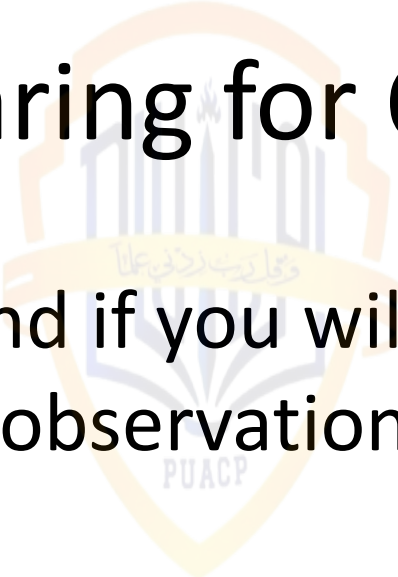
Preparing for Observation

1. Determine the purpose of the observation activity as related to the overall research objectives
2. Determine the population(s) to be observed
3. Consider the accessibility of the population(s) and the venues in which you would like to observe them
4. Investigate possible sites for participant observation

Preparing for Observation

5. Select the site(s), time(s) of day, and date(s), and anticipate how long you will collect participant observation data on each occasion
6. Decide how field staff will divide up or pair off to cover all sites most effectively
7. Consider how you will present yourself, both in terms of appearance and how you will explain your purpose to others if necessary

Preparing for Observation



8. Plan how and if you will take notes during the participant observation activity
9. Remember to take your field notebook and a pen

After Participant Observation

10. Schedule time soon after participant observation to expand your notes
11. Type your notes into computer files using the standard format set for the study

INTER-VIEW



3. Interview

- An INTERVIEW is a data-collection (generation) technique that involves oral questioning of respondents.
- Answers to the questions posed during an interview can be recorded by **writing** them down or by **tape-recording** the responses, or by a combination of both.

High Degree of Flexibility

- When studying sensitive (teenage pregnancy and abortions)
- A list of topics may include:
 - How teenagers started sexual intercourse
 - The actions couples take in the event of unwanted pregnancies
- The investigator should have an additional list of topics ready when the respondent falls silent
 - e.g., when asked about abortion methods used
 - who made the decision and who paid

High Degree of Flexibility

- The sequence of topics should be determined by the flow of discussion.
- It is often possible to come back to a topic discussed earlier in a later stage of the interview.
- The unstructured or loosely structured method of asking questions can be used for interviewing individuals as well as groups of key informants (FGD).

Low Degree of Flexibility

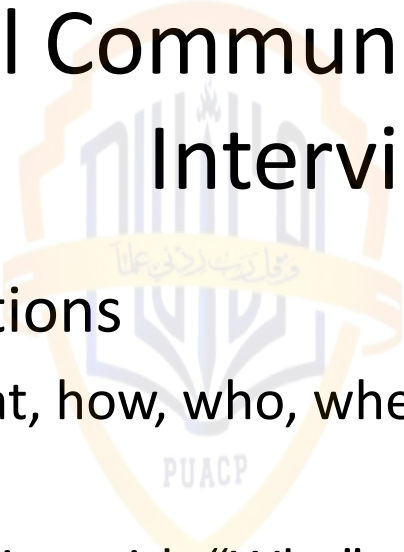
- Less flexible methods of interviewing are useful when the researcher is relatively knowledgeable about expected answers or when the number of respondents being interviewed is relatively large.
- Then **questionnaires** may be used with a fixed list of questions in a standard sequence, which have mainly fixed or pre-categorised answers.

Practical Communication Skills for Interviews

A faint watermark of the PUACP logo is visible in the background. It features a shield-like emblem with Arabic calligraphy at the top and the acronym 'PUACP' at the bottom.

- Good communication should lead to a shared understanding
- Be aware of nonverbal communication (70%)
- Giving constructive feedback
 1. Comment on positive things first
 2. Be constructive
 3. Be specific
 4. Do not give direct or blaming criticism

Practical Communication Skills for Interviews

The logo of the Punjab University of Agriculture, Cholistan (PUACP) is centered in the background. It features a shield with a yellow border, containing a blue and white emblem with Arabic calligraphy and the acronym 'PUACP' at the bottom.

- Ask good questions
 - Start with what, how, who, when, please give an example of
 - Question starting with “Why” make people feel uncomfortable
- Don’t ask a biased and leading or direct questions
- The type of question asked must be adapted to the changing level of trust between interviewer and informant during the interview

Probing

- PROBING is a good questioning skill

Examples

- *Does the child have fever? (Closed question)*
 - Probes How high is the fever?
 - Since when has he had the fever?
- *Have you given the child medicine?*
 - Probes What kinds of medicines have you given?
 - How much? For how long?

Type of Question

- **Hypothetical question:**
 - Suppose you had more money, how would you spend that?
- **Provocative question:**
 - Some say that HIV is the punishment of God, what do you think about that?
- **Ideal question:**
 - Please describe to me what a good delivery would be like.
- **Interpretative question:**
 - You said earlier that you go to healers for diarrhoea, how this related ideas about hot-cold?

Type of Question

- **Experience question:**
 - Could you tell me about your experience of caring for patients with cancer?
- **Feeling question:**
 - How did you feel when the first patient in your care died?
- **Knowledge question:**
 - What services are available for this group of patients?

Type of Question

- **Grand-tour question:**
 - Can you describe a typical day in the community? (to a community midwife)
- **Mini-tour question:**
 - Can you describe what goes on when a women die giving birth?
- **Example question:**
 - Can you give me an example of a difficult delivery?

Type of Question

Type of Question	Purpose
Opening questions	To identify the characteristics that the participants have in common. Participants should be given an opportunity to introduce themselves.
Introductory questions	To introduce the general topic of the discussion, and to stimulate the conversation and improve interaction in the group.
Transition questions	To move the participants into the focus of the discussion.
Key questions	Concern about the focus of the interview.
Ending questions	Give the participants an opportunity to make final statement.
Final questions	Ask the participants to add things they think have not been considered during the discussion.

Prepare for the Interview

- Recruit participants according to the recruitment strategy outlined in the work plan.
- Set up recording equipment and the physical space where interviews will take place.
- Become knowledgeable about the research topic, including anticipating and being prepared to answer any questions participants may have about it.

Prepare for the Interview

- Be reliable. To get participants to take the interview seriously, you need to demonstrate your own commitment. Arrive on time, equipped with the recording equipment, interview guide, and notebooks. Be both mentally and psychologically prepared to conduct the interview. Keep all promises you make to participants.

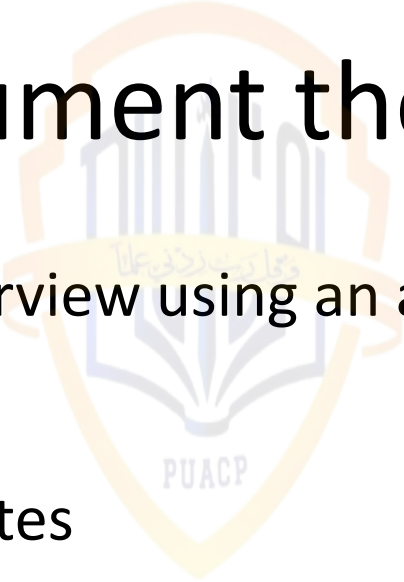
Prepare for the Interview

- Obtain informed consent from each participant before the interview.
- Address all questions or topics listed in the interview guide.
- Ask follow-up questions (some of which may be scripted in the interview guide) in order to elicit participants' complete knowledge and experience related to the research topic.

Prepare for the Interview

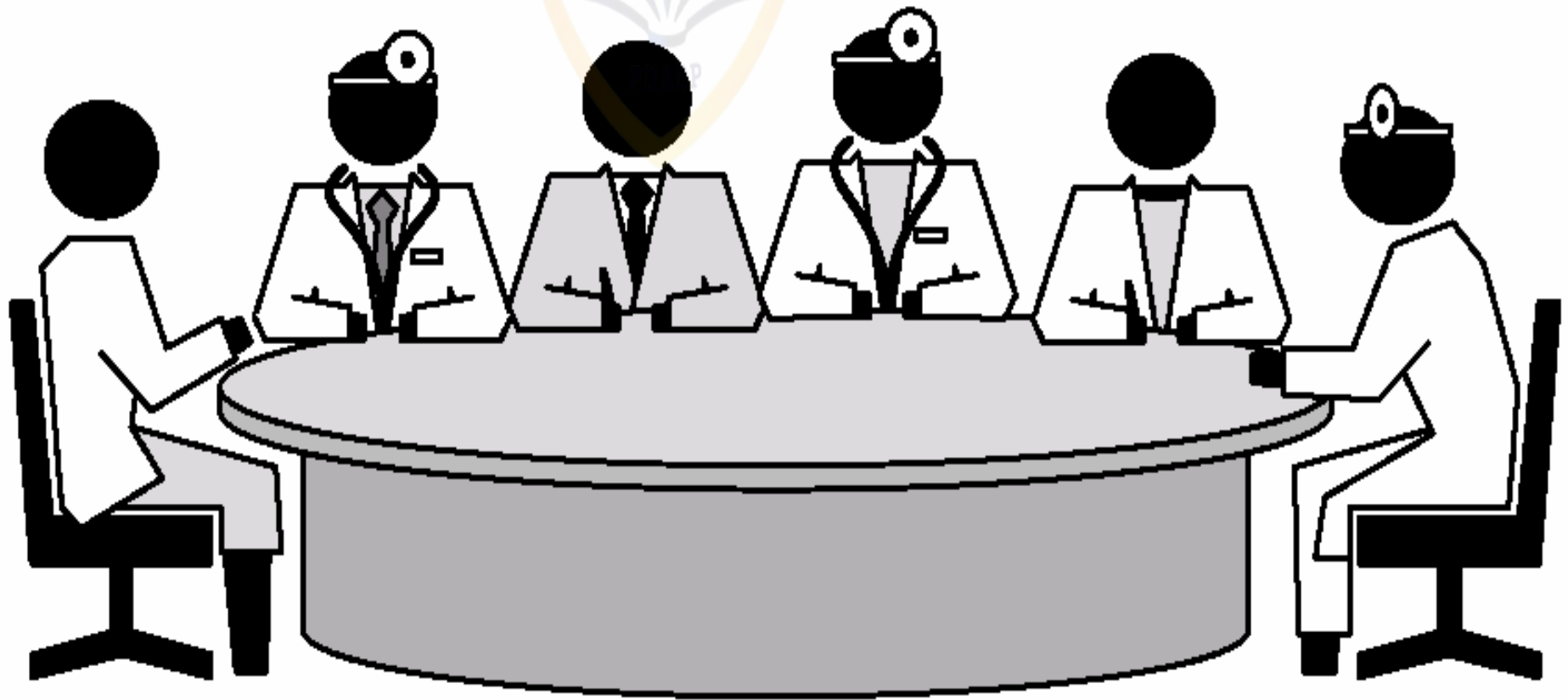
- Probe participants for elaboration of their responses, with the aim of learning all they can share about the research topic.

Document the Interview

The logo of the Palestinian University of Applied Sciences (PUACP) is centered in the background. It features a shield with a yellow border and a blue interior. Inside the shield is a stylized open book with Arabic calligraphy above it. The acronym 'PUACP' is written in blue at the bottom of the shield.

- Record the interview using an audio (and sometimes video) recorder
- Take backup notes
- Observe and document participants' behaviors and contextual aspects of the interview as part of your field notes
- Expand your notes as soon as possible after each interview, preferably within 24 hours, while your memory is still fresh

4. Focus Group Discussion



	Appropriate for	Strength of Method
Interviews	•Eliciting individual experiences, opinions, feelings •Addressing sensitive topics	•Elicits in-depth responses, with differences and contradictions •Gets at interpretive perspective, i.e., the connections and relationships a person sees between particular events, phenomena, and beliefs
Focus Groups	•Identifying group norms •Eliciting opinions about group norms •Discovering variety within a population	•Elicits information on a range of norms and opinions in a short time •Group dynamic stimulates conversation, reactions

Characteristics and Uses of FGD

- A FGD is a qualitative method.
- Its purpose is to obtain in-depth information on concepts, perceptions and ideas of a group.
- A FGD aims to be more than a question-answer interaction.
- The idea is that group members discuss the topic among themselves, with guidance from the facilitator.

FGD Techniques Can be Used To

1- Focus research and develop relevant research hypotheses by exploring in greater depth the problem to be investigated and its possible causes.

example:

- A district health officer had noticed that there were an unusually large number of cases of malnutrition of children under 5 reported from one area in her district.
- Because she had little idea of why there might be more malnutrition in that area she decided to organise 3 FGD (one with leaders, one with mothers and one with health staff from the area).
- She hoped to identify potential causes of the problem through the FGDs and then develop a more intensive study, if necessary.

FGD Techniques Can be Used To

2- Formulate appropriate questions for more structured, larger scale surveys.

example:

- In planning a study of the incidence of childhood diarrhoea and feeding practices, a FGD showed that in the community under study, children below the age of 1 year were not perceived as having 'bouts of diarrhoea' but merely 'having loose stools' that were associated with milestones such as sitting up, crawling, and teething.
- In the questionnaire that was developed after the FGD the concept 'diarrhoea' was therefore carefully described, using the community's notions and terms.

FGD Techniques Can be Used To

3- Help understand and solve unexpected problems in interventions.

example:

- In District X, the recent national (polio) immunisation days (NID) showed widely different coverage's per village (50-90%) and in a number of villages a marked decrease in coverage was observed compared to last year.
- Eight FGD were held with mothers, two in town, three in rural villages with a marked decrease in NID coverage and three in villages with a high coverage throughout.

FGD Techniques Can be Used To

Example (continue)

- It appeared that overall, the concept NID had raised confusion. Most people believed that this mass campaign strengthened the children's immunity against *any* (childhood) disease, including malaria and Respiratory Tract Infections.
- In the villages with a low NID coverage there had been a high incidence of malaria in children immediately after the previous NID campaign and several children died. Mothers therefore believed that the NID campaign was useless.

FGD Techniques Can be Used To

4- Develop appropriate messages for health education programmes and later evaluate the messages for clarity.

example:

- A rural health clinic wanted to develop a health education programme focused on weaning problems most often encountered by mothers in the surrounding villages and what to do about them.
- The FGD could be used for **exploring relevant local concepts** as well as for **testing drafts** when developing the messages.
- The messages should be developed and tested in different socio-economic groups of mothers, as weaning practices may differ with income, means of subsistence and education of the mothers. Also ethnic differences may have to be taken into account.

FGD Techniques Can be Used To

5- Explore controversial topics

example:

- Sexual behaviour is a controversial topic in the sense that males and females judge sexual relations and sexuality often from very different perspectives. Sexual education has to take this difference into account.
- Through FGDs, first with females, then with males, and then with a mixed group to confront both sexes with the different outcomes of the separate discussions (listed on flip charts) it becomes easier to bring these differences in the open. Especially for teenagers, who may have many stereotypes about the other sex or be reluctant to discuss the topic openly (particularly girls), such a 'multi-stage' approach is useful.

Strengths and Limitations of FGD

- A well chosen group, in terms of composition and number, FGDs can be a powerful research tool which provides valuable spontaneous information in a short period of time and at relatively low cost.
- FGD should *not* be used for quantitative purposes, such as the testing of hypotheses or the generalisation of findings for larger areas, which would require more elaborate surveys.

Strengths and Limitations of FGD

- Depending on the topic, it may be **risky** to use FGDs as a **single tool**.
- In group discussions, people tend to centre their opinions on the most common ones, on 'social norms'.
- In reality, opinions and behaviour may be more diverse. Therefore it is advisable to combine FGDs with at least some key informant and in-depth interviews.

Strengths and Limitations of FGD

- In case of very **sensitive topics**, such as sexual behaviour or coping with HIV/AIDS, FGDs may also have their limitations, as group members may hesitate to air their feelings and experiences freely.
- One possible remedy is the selection of participants who do not know each other (e.g., selection of children from different schools in FGDs about adolescent sexual behaviour), while assuring absolute confidentiality.



How To Conduct a FGD

Determine the Purpose

- A FGD can be regarded as a mini-study.
- It therefore requires one or two clear objectives.
- These objectives will guide the research team in the formulation of discussion questions.

Situation Analysis

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- Any FGD requires good knowledge of local conditions.
- There are always differences between community members, for example in education, political power, gender, economic status and ethnic group.
- These differences will be reflected in their perceptions of the problems they suffer from and possible solutions.
- A researcher must be aware of these differences.

Situation Analysis

- The first task of the researchers will be to explore the area and identify possible target groups.
- Interviews with some key informants and a rudimentary situation analysis are then indispensable.
- The situation analysis should preferably be carried out in a participatory way, with representatives of the study population on which the FGD focuses.



Points to be considered when preparing the FGD

A. Recruitment of Participants

- Participants should be roughly of the same socio-economic group or have a similar background in relation to the issue under investigation.
- The age and sexual composition of the group should facilitate free discussion.

B. Selection of Participants

- If you are an outsider, you may have to rely on your key informants for the first selection of participants in FGDs.
- Your key informants to whom you have explained thoroughly the purpose and the process of the FGD might each suggest some individuals who could be invited to a focus group discussion.

B. Selection of Participants

- The key informants may select persons similar to themselves so that you do not get an adequate variety of views in your discussion group.
- So in your explanations be sure to emphasise that you want a group of people that can express a *range* of views, to be able to have a proper discussion.
- Participants in a first FGD may assist to find relevant participants for other groups.
- Another way is to select individuals in a *systematic* way, to try and ensure a range of views. You might, ask every third or fourth person you find. This method might be more suitable in urban areas.

C. Physical Arrangements

- Communication and interaction during the FGD should be encouraged in every way possible.
- Arrange the chairs in a circle. Make sure that there will be no disturbances, sufficient quietness, adequate lighting, etc.
- Try to hold the FGD in a neutral setting which encourages participants to freely express their views.
- A health centre, **for example**, is not a good place to discuss traditional medical beliefs or preferences for other types of treatment.

D. Preparation of a Discussion Guide

- There should be a **written** list of topics to be covered.
- It can be formulated as a series of open-ended questions.
- Guides for different groups gathered to discuss the same subject may vary slightly, depending on their knowledge or attitudes and how the subject should first be explored with them.



Conducting the FGD Session

Functions of the Facilitator

1. Introduce the session
2. Encourage discussion and involvement
3. Deal correctly with sensitive issues
4. Observe non-verbal communication
5. Avoid being placed in the role of expert
6. Control the rhythm of the meeting
7. Take time at the end of the meeting to summarise, check for agreement and thank the participants
8. Summarise the main issues brought up
9. Listen for additional comments and spontaneous discussions after the meeting has been closed

Functions of the Recorder

- The recorder should keep a record of the content of the discussion as well as emotional reactions and important aspects of group interaction.
- Assessment of the emotional tone of the meeting and the group process will enable you to judge the validity of the information collected during the FGD.

Items to be Recorded Include

- Date, time, place
- Names and characteristics of participants
- General description of the group dynamics (level of participation, presence of a dominant participant, level of interest)
- Opinions of participants, recorded as much as possible in their own words
- Emotional aspects
- Vocabulary used – (to use it in developing questionnaires or health education materials)
- Spontaneous relevant discussions during breaks or after the meeting has been closed

Number of Sessions

- It depends upon project needs, resources, and whether new information is still coming from the sessions (Saturation)
- One should plan to conduct at least two FGDs for each sub-group (for example, two for males and two for females)
- Otherwise you have no way of assessing whether the information you get from the first FGD is representative for that group

Duration of Sessions

- A focus group session typically lasts up to an hour and a half.
- Generally the first session with a particular type of group is longer than the following ones because all of the information is new.
- Thereafter, if it becomes clear that all the groups have a similar opinion on particular topics, the facilitator may be able to move the discussion along more quickly to other topics which still elicit new points of view.



End